

15 Examples of Rate Laws in Heterogeneous Catalysis

Why Are We Learning This?

Many catalytic reactions do not follow simple power-law kinetics. Instead, experimentally measured rate laws often contain terms that account for:

Adsorption, Surface Coverage, Competition for Sites, Kinetics of Multiple Reaction Steps

These rate laws provide insight into what is happening on the catalyst surface and help engineers predict reactor performance under realistic operating conditions.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



► Rate Laws ◀
(Kinetics)



Equilibrium
(Reversible Reactions)



Equilibrium Conversion
(Thermodynamic Limitations)



One Reaction: Solving in terms of Reaction Progress



One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)



Catalytic Reactions



Numerical Methods
(Python)



Energy Balances



One Reaction: Non-Isothermal Reactor Design



Multiple Reactions



Reactor Safety



Challenging the Well-Mixed Assumption
(Residence Time Distributions)



Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Learning Objectives

By the end of this lesson, you should be able to:

- Interpret heterogeneous catalytic rate laws.
- Identify the roles of pressure, rate constants, and adsorption constants.
- Explain why adsorption appears in catalytic rate laws.
- Understand how surface coverage influences reaction rate.
- Determine how temperature affects kinetic and adsorption parameters.
- Recognize common features of Langmuir-Hinshelwood-type rate expressions.

Big Idea

Gas-Phase Species



Adsorb on Surface

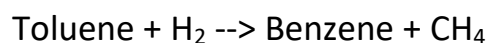


Occupy Catalytic Sites



Affect Reaction Rate

Unlike simple power-law kinetics, heterogeneous catalytic rate laws incorporate both reaction kinetics and adsorption behavior.



$$-r_T = \frac{kP_{\text{H}_2}P_T}{1 + K_BP_B + K_TP_T}$$

Why Do Catalytic Rate Laws Look So Complicated?

Students often expect rate laws to look like:

$$r = kCA \text{ or } r = kCACB$$

However, heterogeneous catalytic reactions occur on catalyst surfaces.

As a result: Adsorption and Availability of Surface Sites must also be considered.

Physical Interpretation

Reactants in Gas Phase



Adsorb on Surface



Compete for Sites



React

This competition for catalytic sites produces the denominator terms that often appear in catalytic rate laws.

$\text{NO} + \text{CO} \rightarrow \text{products}$ (used in diesel exhaust remediation)

$$-r_{\text{NO}} = \frac{kP_{\text{NO}}P_{\text{CO}}}{(1 + K_{\text{NO}}P_{\text{NO}} + K_{\text{CO}}P_{\text{CO}})^2}$$

Dehydration of butanol to butene: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} \rightarrow \text{CH}_3\text{CH}=\text{CHCH}_3 + \text{H}_2\text{O}$

$$-r_{\text{Bu}} = \frac{kP_{\text{Bu}}}{(1 + K_{\text{Bu}}P_{\text{Bu}})^2}$$

What Do These Rate Laws Have in Common?

Although the reactions are different, all three rate expressions share a similar structure.

General Form: Rate = Reaction Term / Adsorption Term

Interpretation:

The numerator describes: Chemical Reaction

while the denominator describes: Competition for Catalytic Sites

Key Insight: More Adsorption $\neq$ Always Faster Reaction

If too many sites become occupied, reactants may actually inhibit the overall reaction.

In these rate laws:

- P are

With units of

$P =$

What Do the Pressures Represent?

In heterogeneous catalysis, reactant concentrations are often expressed as partial pressures rather than molar concentrations.

Physical Meaning

P_A means: Partial Pressure of A

Why Pressure Matters: Increasing pressure generally increases the number of molecules available to adsorb onto the catalyst surface.

Higher Pressure

↓

More Adsorption

↓

Greater Surface Coverage

- k are

With units of

What Does k Represent?

The parameter k is the kinetic rate constant. It describes how rapidly the surface reaction proceeds once the reactants are adsorbed onto the catalyst surface.

Physical Interpretation

Large k



Fast Surface Reaction

Small k



Slow Surface Reaction

Temperature Dependence

The value of k depends strongly on temperature.

Higher Temperature



Larger k



Faster Reaction

As temperature increases, the rate constant typically increases, causing the reaction to proceed more rapidly.

The rate law has units of

k depends on

If you are given a value of k at one temperature and need to compute the value at a new temperature, you can use the

- K are

With units of

They indicate

K depend on

If you are given a value of K at one temperature and need to compute the value at a new temperature, you can use the

Arrhenius Equation and van't Hoff Equation

If the value of k is known at one temperature and must be estimated at another temperature, the Arrhenius equation can be used.

Key Idea

Temperature Changes



Rate Constant Changes



Reaction Rate Changes

The Arrhenius equation connects kinetic behavior to activation energy.

Why This Matters

Higher Temperature



Larger k



Faster Surface Reaction



Higher Overall Rate

In heterogeneous catalytic systems, temperature often affects both:

k



Reaction Kinetics

and

K



Adsorption Equilibrium

which is why catalytic rate laws can change significantly with temperature.

What Does K Represent?

The parameter K is an adsorption equilibrium constant. It describes how strongly a species adsorbs onto the catalyst surface.

Physical Interpretation

Large K



Strong Adsorption

Small K



Weak Adsorption

Why It Matters: Adsorption controls how many catalytic sites are occupied.

Adsorption Strength



Determines K



Determines Surface Coverage



Influences Reaction Rate

A species that adsorbs more strongly generally occupies a larger fraction of the catalyst surface.

Why Does K Depend on Temperature?

Adsorption is an equilibrium process. Like all equilibrium processes, adsorption behavior changes with temperature.

Physical Interpretation

Temperature



Changes Equilibrium Position



Changes K

The adsorption equilibrium constant, K , tells us how strongly a species adsorbs to the catalyst surface. Because adsorption is an equilibrium process, changing the temperature changes the value of K .

Important Distinction

k



Controls Reaction Speed

K



Controls Adsorption Extent

In other words:

k



Kinetics

K



Adsorption Equilibrium

Students frequently confuse these two parameters.

- k determines how rapidly the surface reaction proceeds.
- K determines how strongly molecules bind to the catalyst surface.

Key Insight

Temperature



Changes Both k and K

But In Different Ways. A change in temperature can simultaneously affect the reaction rate constant (k) and the adsorption equilibrium constant (K), which is one reason heterogeneous catalytic kinetics can be more complex than simple power-law rate laws.

Decision Guide: Is Adsorption Helping or Hurting?

Weak Adsorption

Few Occupied Sites



Low Reaction Rate

Moderate Adsorption

Enough Reactants on Surface

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High Reaction Rate

Very Strong Adsorption

Most Sites Filled

↓

Few Empty Sites

↓

Rate May Decrease

Key Insight

Best Performance

↓

Intermediate Surface Coverage

More adsorption is not always better. If adsorption is too weak, insufficient reactant is available on the surface. If adsorption is too strong, catalytic sites become blocked and the reaction rate may decrease.

Toluene + H₂ --> Benzene + CH₄

$$-r_T = \frac{kP_{H_2}P_T}{1 + K_BP_B + K_TP_T}$$

NO + CO → products (used in diesel exhaust remediation)

$$-r_{NO} = \frac{kP_{NO}P_{CO}}{(1 + K_{NO}P_{NO} + K_{CO}P_{CO})^2}$$

Dehydration of butanol to butene: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} \rightarrow \text{CH}_3\text{CH}=\text{CHCH}_3 + \text{H}_2\text{O}$

$$-r_{Bu} = \frac{kP_{Bu}}{(1 + K_{Bu}P_{Bu})^2}$$

Summary and Key Takeaways

Key Ideas

- ✓ Heterogeneous catalytic rate laws include both reaction kinetics and adsorption effects.
- ✓ Partial pressures are often used instead of concentrations.
- ✓ The kinetic constant k describes the speed of the surface reaction.
- ✓ The adsorption constant K describes how strongly species bind to the catalyst surface.
- ✓ The Arrhenius equation is used to determine how k changes with temperature.
- ✓ Adsorption equilibria determine how K changes with temperature.
- ✓ Strong adsorption is not always beneficial because adsorbed species compete for catalytic sites.

Most Important Idea

Reaction Rate

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Depends on Surface Reaction Speed + Surface Coverage

Catalytic rate laws describe both how fast molecules react on the surface and how effectively molecules occupy the catalytic sites where the reaction occurs.